# Task-Relevant Ultrasonic Information Acquisition for Impacted Composites: From Spatial Information to State-Conditioned Sensing

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**Abstract**

Ultrasonic inspection resolves spatially rich internal morphology, yet a downstream engineering decision may require only a task-relevant subset of the available measurements. We propose Task-Relevant Information Acquisition, a two-part framework that first characterizes how information value depends on space, sampling density, task, inspection state, and predictor, and then uses state-conditioned valuation to realize measurements under exact cost. The framework is evaluated on 276 specimens from six experimental domains using strict nested leave-one-domain-out validation, specimen-first inference, and native-raster acquisition accounting. Spatial internal information reduced equal-domain compression-after-impact prediction error by 32.1%, while a registered 25% sparse condition retained 89.9% of the full-field gain. Task-conditioned oracle analyses identified different mechanical and image- reconstruction priorities, and the highest-valued action changed for 70.4% of specimens as evidence accumulated. State-conditioned valuation reduced next-action regret relative to a static reference in five of six held-out domains. Matched information-source controls and component substitutions then separated state dependence, content attribution, valuation, and bounded planning effects. These results support a task-oriented progression from spatial information characterization to cost-constrained sensing, while retaining explicit boundaries between retrospective opportunity, legal-state valuation, and deployable decision realization.

*Keywords:* adaptive ultrasonic inspection, compression after impact, task-relevant information, state-conditioned sensing, cost-constrained acquisition

## 1. Introduction

Ultrasonic inspection exposes spatially distributed internal damage in impacted carbon-fiber-reinforced polymer (CFRP) structures. A completed scan, however, is an information-rich field rather than an engineering decision. When the downstream objective is compression-after-impact (CAI) assessment, the acquisition problem is to determine which measurements justify their cost because they improve the task-relevant estimate.

Post-impact prediction, sparse ultrasound, adaptive inspection, and task-driven experimental design provide the foundations for this problem. Surface and image-resolved predictors support residual-capacity assessment [1, 2, 3, 4]; limited-data ultrasound segmentation and adaptive reconstruction or path selection address distinct data and measurement constraints [5, 6, 7]; and value-of- information methods connect sensing choices to downstream utility [8, 9, 10].

Existing work does not yet provide an integrated account of how useful ultrasonic information is distributed across a specimen, how its value changes as evidence accumulates, and how local value is converted into a legal bounded measurement set. Full-field predictive accuracy does not identify where a limited budget should be spent. Reconstruction quality may prioritize a different region from CAI loss, and a static priority map cannot represent a value target that changes with the current inspection state.

We therefore ask two linked questions. RQ-A asks how task-relevant ultrasonic information is structured under limited sensing for downstream CAI assessment. RQ-B asks how the resulting information structure can support state- conditioned, task-oriented acquisition under exact cost. The proposed Task- Relevant Information Acquisition framework answers RQ-A through Information Characterization and RQ-B through State-Conditioned Task-Oriented Acquisition under one causal acquisition contract.

The evidence program contains 276 paired specimens from six experimental domains. It uses strict nested leave-one-domain-out evaluation, source-only selection, specimen-first uncertainty estimation, and equal-domain aggregation. The outer-domain policy endpoint is kept separate from diagnostic analyses, and oracle or teacher quantities are used only to characterize opportunity or a prediction target unavailable at decision time.

The contribution is threefold. First, the framework characterizes task-relevant information from spatial enrichment and sparse recoverability through task-, state-, and predictor-conditioned value. Second, it formulates legal partial-state valuation and matched source controls that distinguish state dependence from content attribution. Third, it connects estimated value to cost-constrained set realization and evaluates the resulting evidence chain with exact native-raster cost, synchronized bootstrap contrasts, and held-out-domain evaluation.

## 2. Related Work

### *2.1. Post-impact ultrasonic information and residual-capacity assessment*

Low-velocity impact can produce distributed internal damage with limited surface evidence, motivating paired surface, ultrasonic, and CAI assessment. Surface profiles have supported internal-damage and residual-strength prediction [1, 2], while image- resolved analyses show why spatial morphology should not be reduced to a single projected extent [3]. The paired public releases used here contain specimen metadata, surface profiles, ultrasonic C-scans, and CAI measurements [11, 12].

Full C-scan images can directly predict impact energy and CAI strength with a convolutional model [4]. This establishes the predictive value of the completed field. Task-relevant acquisition addresses the preceding decision: how much of that field should be measured, which location should be selected next, and which parts of the current state can support that choice before the scan is complete.

### *2.2. Sparse and adaptive ultrasonic acquisition*

Learning-based ultrasonic damage segmentation has used augmentation and synthetic examples when available scan datasets are small [5]. Separately, adaptive composite inspection has combined damage-guided sampling with masked reconstruction [6], and robotic ultrasound has selected successive physical locations with Bayesian optimization over damage-probability or novelty fields [7]. These approaches establish sparse and sequential acquisition as practical design patterns.

Their objectives are not interchangeable. Reconstruction-oriented sampling rewards measurements that improve a recovered field, whereas task-oriented sampling rewards measurements that reduce downstream CAI loss. A mechanically useful measurement need not minimize global image error. Comparing both objectives under the same legal action and exact-cost contract makes this distinction observable without translating a digital sampling fraction into scanner time.

### 2.3. *Task-relevant information acquisition formulation*

Task-driven experimental design chooses compact measurements for a specified downstream objective [8]. Ultrasonic value-of- information criteria similarly trade diagnostic benefit against sensor number or placement cost [10]. In partially observable asset management, observation value changes with the belief state and future opportunities [9]; the same dependency arises during a partial C-scan.

Task-Relevant Information Acquisition combines these principles at the level of the current specimen and legal measurement state. Its distinctive operational sequence is to characterize task value, estimate it from current evidence, test its information source with matched controls, and realize a bounded set under exact cost. Retrospective opportunity, deployable valuation, and policy performance are therefore evaluated as related but non-equivalent quantities.

## 3. Task-Relevant Information Acquisition Framework

### 3.1. *Task-Relevant Information Characterization*

Let specimen $i$ belong to domain $d(i)$, let $y_i$ be its damaged-to-intact CAI-strength ratio, and let $\mathcal{I}_{i,t}$ denote information legally available after acquisition step $t$. This state contains pre-inspection context, acquired coordinates and values, action history, and exact cost, but excludes unrevealed ultrasonic values and $y_i$. For downstream predictor $f$ and loss $\ell$, the current risk is

$$\mathcal{R}_f(\mathcal{I}_{i,t}) = \ell(y_i, f(\mathcal{I}_{i,t}))\,, \tag{1}$$

and the realized value of legal candidate $X \in \mathcal{A}(\mathcal{I}_{i,t})$ is

$$U_f(X \mid \mathcal{I}_{i,t}) = \mathcal{R}_f(\mathcal{I}_{i,t}) - \mathcal{R}_f(\mathcal{I}_{i,t} \cup X). \tag{2}$$

A positive value means that revealing $X$ reduces loss for predictor $f$ at this state. The predictor index and state are essential: value is neither an intrinsic coordinate property nor a universal mechanical map.

Information Characterization varies the relevant dependency while holding the cohort, held-out domains, loss, and cost logic fixed. Matched scalar-to-spatial comparisons test information enrichment, a registered sparse protocol tests recoverability, candidate comparisons test spatial heterogeneity, common-cost mechanical and reconstruction objectives test task conditioning, and repeated states and predictors test state and predictor conditioning. A retrospective oracle may maximize Eq. (2) using counterfactual measurements and true outcomes; it characterizes opportunity and remains non-deployable.

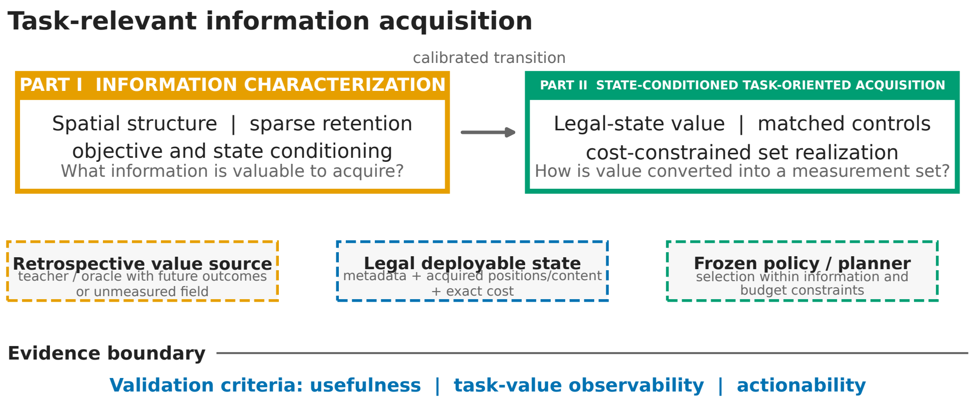


Figure 1: Task-Relevant Information Acquisition. Part I characterizes which spatial measurements matter for a downstream task and how their value changes with state and predictor. Part II estimates value from legal partial state, tests its source with matched controls, and realizes a measurement set under exact cost. Oracles and controls are evidence instruments rather than proposed methods.

### *3.2. State-Conditioned Measurement Valuation*

A scorer $g_\theta$ is restricted to the current legal state, candidate descriptor, and incremental cost:

$$\widehat{U}_{i,t,X} = g_\theta(\mathcal{I}_{i,t}, X, c(X \mid \mathcal{I}_{i,t}))\,. \tag{3}$$

For each outer target domain, teacher predictors, state encoders, scorer fitting, and selection use source domains only. Strict out-of-fold teacher values compare current and counterfactually updated CAI error. The held-out domain is queried only after every fitting and selection choice is fixed.

Matched controls determine what Eq. (3) has learned. A static reference removes state conditioning. The acquired-position/history control inherits the exact coordinates, history, and cost while removing measured values. A reconstruction control derives information from the legal partial field, and a shuffled-content control preserves recipient positions and budget while replacing values with source-separated donor content. Improvement over the static reference supports state dependence; attribution to measured content additionally requires favorable matched-control evidence.

### 3.3. Cost-Constrained Task-Oriented Acquisition

The task-oriented selector converts estimated value into a legal measurement. A one-step value-per-cost policy is

$$\pi(\mathcal{I}_{i,t}) = \arg\max_{X\in\mathcal{A}(\mathcal{I}_{i,t})} \frac{\widehat{U}_{i,t,X}}{c(X \mid \mathcal{I}_{i,t})}, \tag{4}$$

subject to hierarchical refinement and remaining budget. Bounded lookahead uses the same reveal contract. Only coordinates implied by a selected action are materialized, the state is rebuilt after reveal, and every transition is charged by unique newly observed native-raster locations.

For effective specimen budgets $x_{i,1} < \cdots < x_{i,K}$ and CAI errors $e_i(x_{i,k})$, trajectory quality is

$$\mathrm{AUEBC}_i = \frac{1}{x_{i,K} - x_{i,1}} \sum_{k=1}^{K-1} \frac{e_i(x_{i,k}) + e_i(x_{i,k+1})}{2}(x_{i,k+1} - x_{i,k}), \tag{5}$$

the budget-span-normalized trapezoidal mean error over the observed effective-budget range. Lower values are preferred. Component substitutions separate valuation and planning effects, while reachable-set comparisons test whether local value is realized as an effective bounded set.

## 4. Multi-Domain CFRP Experimental Design

### 4.1. Dataset and Information Representations

The study uses 276 CAI-complete CFRP specimens from six released experimental datasets [11, 12]. The domains contain 45, 49, 43, 59, 42, and 38 specimens and remain indivisible evaluation units. Each specimen has metadata, surface observations, an ultrasonic RGB C-scan, and a damaged-to-intact CAI-strength ratio. Source manifests bind specimen and domain identifiers to native raster shape and content hashes.

Information representations span scalar summaries, the selected spatial full field, a registered 25% bilinear sparse field, and sequential legal states. The sparse condition is sampled on a normalized raster before the frozen field pathway. Sequential acquisition partitions each native scan into an 8×8 cell grid with three refinement levels. Candidate descriptors store cell position, level transition, added native locations, native count, and remaining budget. Mechanics utility uses CAI loss; image utility uses the registered normalized-RGB-MSE reconstruction objective.

Table 1: Case study and evaluation protocol.

| Item | Value |
|---|---|
| Cohort | 276 physical specimens across 6 experimental domains; domain counts 45, 49, 43, 59, 42, 38 |
| Information representations | Specimen metadata and surface observables; full ultrasonic C-scan field; 25% normalized-raster sparse field; current partial state |
| Actions and cost | 8 x 8 cells with three acquisition levels; cost is exact unique native-raster locations divided by native count, with no scanner-time equivalence |
| Information boundary | Deployable state contains metadata, acquired positions and content, legal actions, and exact cost; teacher and oracle information is retrospective and non-deployable |
| Evaluation | Strict nested leave-one-domain-out (LODO); outer-domain outcomes are excluded from training, selection, and calibration |
| Statistics | physical specimen first and held-out domain second; equal-domain aggregation with synchronized within-domain bootstrap; repeated computational records are not independent samples |

### *4.2. Causal Acquisition Protocol*

Every decision begins from a geometry-neutral scout and a legal partial state. Legal inputs comprise metadata, acquired positions, measured current content, measurement levels, action history, available actions, and exact current cost. Unrevealed scan content and held-out CAI outcomes are forbidden. Teacher and oracle calculations may access counterfactual quantities only after the deployable information boundary has been recorded and only for source-domain training or retrospective evaluation.

An action refines one cell by one level. Its cost is the number of unique newly revealed native-raster coordinates; already measured locations add zero cost, and an action exceeding the cap is illegal. Registered checkpoints are 3.125%, 6.25%, 9.375%, 12.5%, 18.75%, and 25%. Initial scouts are 1.5625% except where native geometry requires 3.125%. No result is translated into physical scanner time.

### *4.3. Held-Out-Domain Evaluation and Statistical Analysis*

All headline estimates use nested leave-one-domain-out evaluation. Each domain serves once as the outer target; normalization, fitting, checkpoints, hyperparameters, fallbacks, and policy selection use the other five domains. When inner selection is needed, source domains remain intact [13, 14]. The physical specimen is the first statistical unit. Candidate actions, checkpoints, and seeds from one specimen are repeated computational records, not independent samples.

Contrasts use synchronized within-domain specimen bootstrap resampling followed by equal-domain aggregation, preserving the nested specimen/domain structure [15]. The registered scalar-to-field family retains its familywise simultaneous interval, while other analyses retain their registered paired intervals. With six domains, direction counts accompany aggregate effects. Native-raster cost is exact digital coverage, not travel time, coupling time, or industrial inspection cost.

## 5. Experimental Results and Discussion

### *5.1. From Spatial Information to State-Conditioned Task Value*

#### *5.1.1. Spatial information and sparse recoverability*

The matched spatial field reduced equal-domain CAI-ratio MAE from 0.18920 to 0.12849, a reduction of 0.06072 (32.1%; familywise 95% CI 0.00664–0.15364), and improved five of six held-out domains. Relative to the metadata-and-surface reference, the reduction was 0.05963 (31.7%; familywise 95% CI 0.00722–0.14842), again favorable in five domains. Spatial internal information therefore adds CAI-relevant signal beyond the lower-dimensional representations under the matched predictor family.

At 25%, the registered bilinear sparse condition reduced error relative to the surface reference by 0.05361 (95% CI 0.00561–0.13544), improved five of six domains, and retained 89.9% of the full-field gain. Its MAE remained 0.00602 above the selected full field (95% CI 0.00173–0.01083), with the sparse condition higher in all six domains. Sparse native-raster observation thus retains most, but not all, of the registered spatial gain.

#### *5.1.2. Task-conditioned spatial measurement value*

One-shot mechanical-oracle CAI AUEBC was 0.013458, compared with 0.017363 for uniform acquisition and 0.017188 for the one-shot reconstruction oracle. The mechanical improvements were 0.00391 (95% CI 0.00280–0.00502) and 0.00373 (95% CI 0.00284–0.00469), respectively, favorable in all six domains. The one-shot oracle retained 56.8% of sequential-oracle headroom, showing that spatial opportunity is heterogeneous while remaining retrospective.

The preferred locations also depended on task. Under the same exact-cost contract, the mechanics oracle improved CAI AUEBC over the reconstruction oracle by 0.04862 (95% CI 0.04527–0.05205). In the reverse task, the reconstruction oracle improved normalized RGB image MSE by $5.503 \times 10^{-4}$ (95% CI $5.006 \times 10^{-4}$–$6.063 \times 10^{-4}$). Mechanics and image objectives therefore induce distinct retrospective priorities.

#### *5.1.3. State- and predictor-conditioned measurement value*

Between the initial state and the 18.75% checkpoint, the strict out-of-fold teacher changed its best action for 70.4% of specimens. Spearman agreement with initial action values was 0.405, top-five overlap was 0.307, and the descriptive dynamic opportunity was 0.00531. Measurement value therefore changes materially as legal evidence accumulates.

Value also retained the downstream-predictor index. Full-state equal-domain strict-OOF MAE was 0.08964 for Ridge, 0.08618 for Huber, and 0.15067 for the shallow MLP. Action-value rank agreement was 0.762 (95% CI 0.699–0.821) between the comparably performing low-complexity predictors and 0.116 (95% CI 0.069–0.164) between Ridge and the substantially less accurate shallow MLP. We retain the predictor index $f$ because the experiment does not determine how much value-map variation remains among equally accurate but structurally distinct predictors.

### *5.2. State-Conditioned Task-Oriented Acquisition*

#### *5.2.1. State-conditioned valuation improves next-action estimation*

At 18.75%, dynamic real minus static was $-0.001260$ in one-step value regret (95% CI $-0.002123$ to $-0.000444$), favorable in five of six held-out domains. Because lower regret is better, the conditional real-state scorer improved next-action estimation relative to the registered static reference.

The static pre-acquisition score had equal-domain Spearman correlation $-0.0196$ with strict-OOF action values (95% CI $-0.0591$–$0.0195$), with the favorable direction in three domains. The interval includes zero; the result defines a tested reference rather than an impossibility claim.

#### *5.2.2. Information-source and component decomposition*

Real partial measurements changed CAI prediction relative to the initial state: at 25%, real-state MAE changed by $-0.000731$ (95% CI $-0.001184$ to

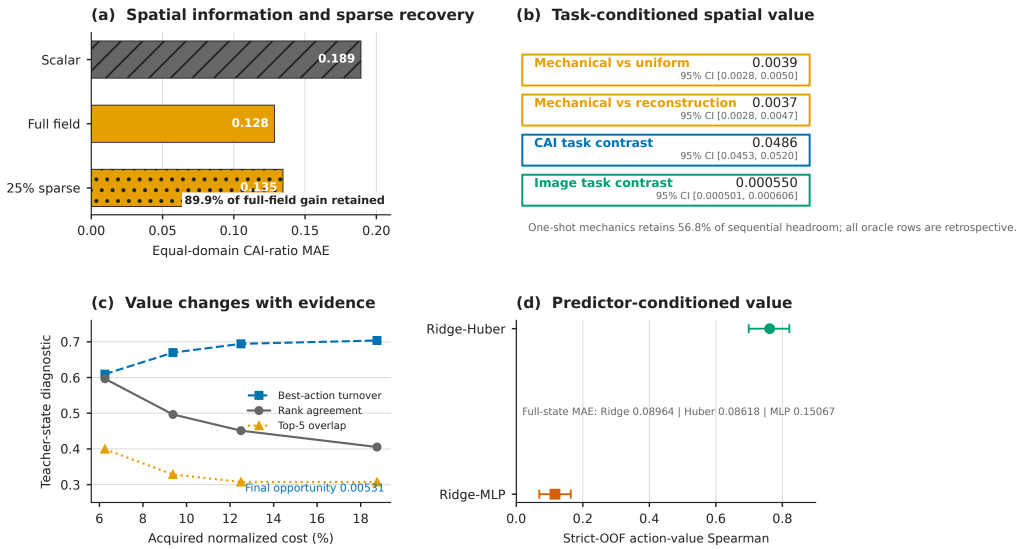


Figure 2: Task-relevant information characterization. (a) Spatial internal information improves CAI prediction and the registered sparse field retains most of the gain. (b) Exact-cost oracle comparisons show heterogeneous and task-conditioned spatial value. (c) Teacher priorities change as evidence accumulates. (d) Rank agreement depends on the downstream predictor.

$-0.000250$). The matched controls delimit its source. Real minus acquired-position/history MAE was 0.01740 (95% CI 0.00890–0.02582), and real minus reconstruction MAE was 0.03419 (95% CI 0.02450–0.04384); real content was favorable in one of six domains for each contrast. The position control inherits the actual acquisition history and is not geometry alone.

Dynamic real minus shuffled-content regret was $2.328 \times 10^{-4}$ (95% CI $6.060 \times 10^{-5}$–$4.180 \times 10^{-4}$), favorable for real content in one domain. Together, these controls show state dependence while restricting a stronger attribution to specimen-specific measured content for the registered representation.

Component substitutions localized downstream headroom. Replacing learned valuation with privileged values improved CAI AUEBC by $4.979 \times 10^{-5}$ (95% CI $1.845 \times 10^{-5}$–$8.105 \times 10^{-5}$). A bounded learned-planning substitution improved AUEBC by $3.164 \times 10^{-6}$ (95% CI $1.411 \times 10^{-7}$–$6.250 \times 10^{-6}$), while true-value planning improved it by $1.117 \times 10^{-4}$ (95% CI $9.898 \times 10^{-5}$–$1.248 \times 10^{-4}$). These retrospective substitutions separate valuation and planning effects without defining deployable modules.

Table 2: Task-relevant acquisition results across the six-stage evidence chain.

| Stage | Question | Headline evidence | Scope boundary |
|---|---|---|---|
| Spatial information and sparse recoverability | Does spatial ultrasonic information improve CAI estimation, and how much of the gain survives sparse observation? | 32.1% MAE reduction; 89.9% of full-field gain retained | Matched confirmatory field and retrospective 25% normalized-raster sampling; normalized cost is not scanner time |
| Task-conditioned spatial measurement value | Is measurement opportunity spatially heterogeneous and conditioned on the downstream task? | Mechanical vs uniform 0.0039; CAI task contrast 0.0486; image task contrast 5.503e-04 | Oracle rows and cross-task priorities are retrospective; learned-policy specificity is outside this test |
| State- and predictor-conditioned measurement value | Does measurement value change with accumulated evidence and the downstream predictor? | Best-action turnover 70.4%; value-rank agreement 0.762 vs 0.116 | Teacher targets are retrospective; the shallow MLP is less accurate, and equal-accuracy structurally distinct predictors remain unresolved |
| State-conditioned valuation | Can legal-state conditioning improve next-action valuation over a static scorer? | Dynamic - static regret -0.0013; static Spearman -0.0196 | Same legal actions and exact cost at 18.75%; this does not uniquely attribute the gain to accumulated measured content |
| Information-source and component decomposition | Which legal-state sources and bounded components account for the remaining acquisition headroom? | Real - history 0.0174; real - reconstruction 0.0342; valuation substitution 4.979e-05 | Matched controls preserve position history and exact cost; component substitutions are retrospective and do not identify a representation bottleneck |
| Cost-constrained set realization | How closely does bounded set realization approach a retrospective joint near-oracle set? | Greedy regret 1.207e-04; beam-4 regret 1.127e-04 | Two-action reachable-pool diagnostic at 6.25%; the near-oracle is non-deployable and does not establish arbitrary-horizon regret |

### *5.2.3. Cost-constrained set realization*

Within the registered two-action reachable pool at 6.25%, current greedy selection had regret $1.207\times10^{-4}$ relative to a retrospective joint near-oracle set (95% CI $1.033\times10^{-4}$–$1.386\times10^{-4}$). Beam search with width four reduced the point regret to $1.127\times10^{-4}$, with a positive interval ($9.485\times10^{-5}$–$1.308\times10^{-4}$). Local value must therefore be realized as a bounded set; the diagnostic does not extend to arbitrary horizons.

A final system-level diagnostic examined one supervised state-conditioned implementation against the static reference. Equal-domain CAI AUEBC was 0.125053 for the learned implementation and 0.124992 for the reference; reference minus learned was $-6.114\times10^{-5}$. The observed direction favors the reference, so this endpoint is an implementation boundary rather than the performance definition of Task-Relevant Information Acquisition.

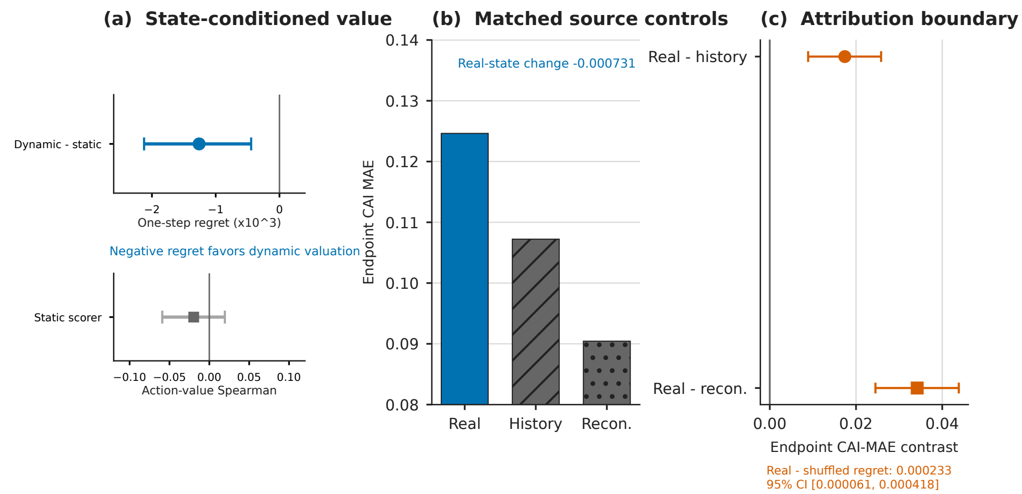


Figure 3: State-conditioned valuation and information-source controls. (a) Dynamic valuation improves next-action regret over the static reference. (b) Real partial state changes CAI prediction, while acquired-position/history and reconstruction controls delimit measured-content attribution. (c) The shuffled-content contrast preserves the same causal boundary.

### *5.3. Engineering Interpretation*

Part I shows why acquisition should be organized around information value rather than scan completeness. Spatial morphology carries CAI-relevant signal, most of the registered gain survives sparse observation, and exact-cost oracle comparisons show that preferred locations depend on the downstream task. The changing teacher priority further rules out a single fixed value map for all inspection states.

Part II shows how that structure enters a decision. State-conditioned valuation improves over the static reference, but matched controls determine how narrowly the state signal can be attributed. Component and reachable-set analyses then separate value estimation from the planner that converts local scores into a budgeted measurement set. The framework therefore treats characterization, valuation, source attribution, and realization as connected engineering-informatics operations.

The present study uses exact native-raster acquisition cost and six CFRP experimental domains. Measurement value is defined relative to the downstream CAI predictor and the registered action space. Accordingly, the reported effects apply to these specimens, targets, costs, and legal sensing

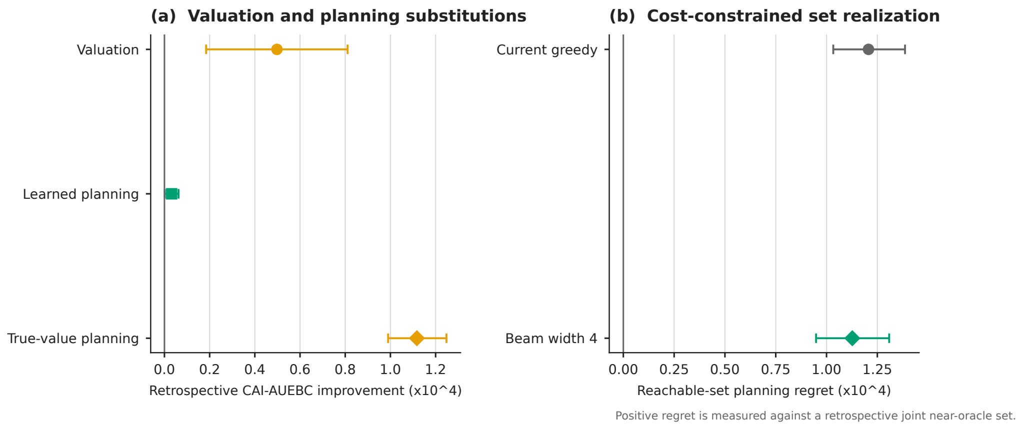


Figure 4: Valuation, planning, and set realization. (a) Controlled substitutions separate valuation and bounded-planning effects. (b) Greedy and beam-4 selection retain positive regret relative to a retrospective joint set within the registered reachable pool. The figure contains component and set-realization evidence only.

decisions. Extending the framework to new materials, sensing hardware, or physical inspection-cost models will require prospective validation under the corresponding measurement process.

## 6. Conclusions

Task-Relevant Information Acquisition connects two scientific parts. Information Characterization establishes spatial enrichment, sparse recoverability, task- conditioned priorities, and state- and predictor-conditioned value. In the six-domain CFRP study, the matched spatial field reduced CAI-ratio error by 32.1%, the sparse field retained 89.9% of that gain, and the best teacher action changed for 70.4% of specimens as evidence accumulated.

State-Conditioned Task-Oriented Acquisition converts that structure into legal partial-state valuation and cost-constrained sets. Dynamic valuation improved next-action estimation over the static reference, matched controls bounded the licensed information-source claim, and component and planning analyses localized distinct realization headroom. The conclusions remain tied to the registered tasks, representations, action space, held-out domains, and exact native-raster accounting.

The engineering contribution is a traceable route from spatial information to state-conditioned sensing: characterize what is useful, estimate value from what is legally known, test where that signal comes from, and realize it under cost. The proposed framework shifts ultrasonic inspection from measuring the complete field by default toward measuring the information that matters for the downstream engineering task.

## Data and code availability

The paired experimental datasets analyzed in this study are public sources cited in Section 4. An anonymized reproducibility package accompanies the submission and contains canonical metrics, source hashes, figure and table sources, supplemental machine-readable records, and a deterministic source archive. Repository identity and archival metadata will be declared after review.

## Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this work, the authors used OpenAI Codex to assist with evidence auditing, deterministic analysis-code generation, LaTeX drafting, and language editing. The authors reviewed and validated all generated content against the original machine-readable artifacts and take full responsibility for the manuscript.

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